

Technical University of Denmark

SPRING, 2025
Campus: Lyngby

MODEL-BASED MACHINE LEARNING

Mini-test 1 — Solutions

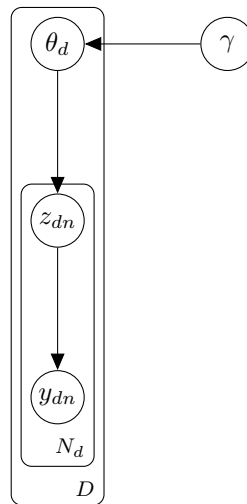
(Time allowed: 60 MINUTES)

Name: _____

Student ID: _____

Part 1

Consider the following probabilistic graphical model (PGM):



1. (10%) Write the factorization of the joint distribution according to the PGM described above.

Solution:

$$p(\gamma, \{\theta_d\}, \{z_{dn}\}, \{y_{dn}\}) = p(\gamma) \prod_{d=1}^D p(\theta_d \mid \gamma) \prod_{n=1}^{N_d} p(z_{dn} \mid \theta_d) p(y_{dn} \mid z_{dn}).$$

2. (10%) Which of the following expressions corresponds to a valid expression for $p(y_{dn} \mid \theta_d)$? Only one option is correct.

- (a) $\sum_{z_{dn}} p(y_{dn} \mid z_{dn}) p(z_{dn} \mid \theta_d)$
- (b) $\sum_{z_{dn}} p(y_{dn} \mid z_{dn}, \theta_d)$
- (c) $p(y_{dn} \mid z_{dn}) p(z_{dn} \mid \theta_d)$
- (d) $\sum_{\theta_d} p(y_{dn} \mid z_{dn}) p(\theta_d)$

Solution: Correct option: (1).

We marginalize the latent z_{dn} :

$$p(y_{dn} \mid \theta_d) = \sum_{z_{dn}} p(y_{dn}, z_{dn} \mid \theta_d) = \sum_{z_{dn}} p(y_{dn} \mid z_{dn}, \theta_d) p(z_{dn} \mid \theta_d).$$

From the PGM, $y_{dn} \perp\!\!\!\perp \theta_d \mid z_{dn}$, hence $p(y_{dn} \mid z_{dn}, \theta_d) = p(y_{dn} \mid z_{dn})$, giving option (1).

3. (15%) Classify the following statements as “True” or “Cannot prove” using the d-separation algorithm:

- (a) $\gamma \perp\!\!\!\perp y_{dn} \mid \theta_d$
- (b) $z_{d1} \perp\!\!\!\perp z_{d2} \mid \gamma$

Solution: Using d-separation:

- (a) **True.** Conditioning on θ_d blocks the chain $\gamma \rightarrow \theta_d \rightarrow z_{dn} \rightarrow y_{dn}$.
- (b) **Cannot prove.** The path $z_{d1} \leftarrow \theta_d \rightarrow z_{d2}$ remains open; conditioning on γ does not block it.

Part 2

Consider the following generative process:

1. Draw $\mu \sim \mathcal{N}(0, 1)$
2. Draw $\sigma^2 \sim \text{Inverse-Gamma}(1, 1)$
3. For each region $r \in \{1, \dots, R\}$:
 - (a) Draw region-specific intercept $\alpha_r \sim \mathcal{N}(\mu, \sigma^2)$
 - (b) For each individual $i \in \{1, \dots, N_r\}$:
 - i. Draw $y_{ri} \sim \text{Bernoulli}(\text{Sigmoid}(\alpha_r + \beta^T \mathbf{x}_{ri}))$

1. (10%) Classify the following statements as True (T) or False (F):

- (a) The model is hierarchical.
- (b) Observations within the same region are conditionally independent given α_r .
- (c) Observations within the same region are marginally independent.
- (d) The model is a finite mixture model.

Solution:

- (a) **T**. Global hyperparameters (μ, σ^2) and region-level latent α_r form a hierarchy.
- (b) **T**. Given α_r (and fixed β), the y_{ri} are independent across i .
- (c) **F**. Marginally, observations share α_r , inducing dependence.
- (d) **F**. No discrete latent assignment selecting among multiple components for each observation.

2. (5%) Suppose posterior (Bayesian) inference yields a very large variance for σ^2 . Which interpretation is most appropriate?

- (a) Regions are very similar.
- (b) There is strong pooling across regions.
- (c) There is large heterogeneity between regions.
- (d) The model is misspecified.

Solution: Correct option: (3). Large σ^2 implies large variation in α_r across regions (high heterogeneity / weak pooling).

Part 3

Suppose data $\{x_n\}_{n=1}^N$ are generated according to:

1. Draw $\pi \sim \text{Beta}(a, b)$
2. For each n :
 - (a) Draw $z_n \sim \text{Bernoulli}(\pi)$
 - (b) If $z_n = 0$, draw $x_n \sim \text{Poisson}(\lambda)$
 - (c) If $z_n = 1$, draw $x_n \sim \text{NegBin}(r, p)$

1. (5%) Which of the following expressions corresponds to the marginal likelihood $p(x_n \mid \pi, \lambda, r, p)$? Only one option is correct.
 - (a) $\pi \text{Poisson}(x_n \mid \lambda) + (1 - \pi) \text{NegBin}(x_n \mid r, p)$
 - (b) $(1 - \pi) \text{Poisson}(x_n \mid \lambda) + \pi \text{NegBin}(x_n \mid r, p)$
 - (c) $\sum_{z_n} p(z_n \mid \pi)$
 - (d) $\text{Poisson}(x_n \mid \lambda) \text{NegBin}(x_n \mid r, p)$

Solution: Correct option: (2).

$$p(x_n \mid \pi, \lambda, r, p) = \sum_{z_n \in \{0,1\}} p(x_n \mid z_n, \lambda, r, p) p(z_n \mid \pi) = (1 - \pi) \text{Poisson}(x_n \mid \lambda) + \pi \text{NegBin}(x_n \mid r, p).$$

2. (5%) Which of the following is a key difference between hierarchical and mixture models?
 - (a) Mixture models contain latent discrete assignments.
 - (b) Hierarchical models cannot contain latent variables.
 - (c) Mixture models are always dynamic.
 - (d) Hierarchical models require Gaussian priors.

Solution: Correct option: (1).

3. (10%) Classify the following statements as True (T) or False (F).
 - (a) Conditioned on π , the latent variables z_n are independent.
 - (b) If π is known and fixed, observing $z_1 = 1$ does not change the probability that $z_2 = 1$.
 - (c) If π has a Beta prior and is not observed, then observing $z_1 = 1$ increases the posterior probability that $z_2 = 1$.
 - (d) Conditioned on the latent variables z_n , the observations x_n are independent.

Solution:

- (a) **T.** Given π , we have $z_n \stackrel{iid}{\sim} \text{Bernoulli}(\pi)$.
- (b) **T.** If π is fixed/known, z_1 provides no information about z_2 beyond π .
- (c) **T.** With $\pi \sim \text{Beta}(a, b)$, observing $z_1 = 1$ increases the posterior mean of π , hence increases $\mathbb{P}(z_2 = 1 \mid z_1 = 1)$.
- (d) **T.** Given z_n , each observation x_n is generated independently from either a Poisson or Negative Binomial distribution, and there are no remaining dependencies between observations.

Part 4

Consider the following dynamic model:

1. Draw $\beta \sim \mathcal{N}(\mathbf{0}, 10\mathbf{I})$
2. Draw $\alpha_1 \sim \mathcal{N}(0, 1)$
3. For $t = 2, \dots, T$:
 - (a) Draw $\alpha_t \sim \mathcal{N}(0.7\alpha_{t-1}, 1)$
4. For each t :
 - (a) Draw $y_t \sim \text{Poisson}(\exp(\alpha_t + \beta^T \mathbf{x}_t))$

1. (10%) Classify the following statements as True (T) or False (F).

- (a) The process $\alpha_t \sim \mathcal{N}(0.7\alpha_{t-1}, 1)$ defines an autoregressive process of order 1 (AR(1)) on α_t .
- (b) The observations y_t and y_{t-1} are conditionally independent given α_t , α_{t-1} and β .
- (c) The observations y_t and y_{t-1} are conditionally independent given α_{t-1} and β only.
- (d) Marginally (i.e., without conditioning on $\alpha_{1:T}$ and β), the sequence $\{y_t\}$ is independent over time.

Solution:

- (a) **T**. α_t depends only on α_{t-1} (order-1 Markov).
- (b) **T**. Given α_t , α_{t-1} and β , the observations are generated independently at each time step.
- (c) **F**. y_t depends on α_t , which is not conditioned on.
- (d) **F**. Temporal dependence is induced through the latent chain $\alpha_{1:T}$.

2. (10%) Write the factorization of the joint distribution according to the generative process described above.

Solution: Let $\alpha_{1:T} = \{\alpha_1, \dots, \alpha_T\}$ and $y_{1:T} = \{y_1, \dots, y_T\}$. Then

$$p(\beta, \alpha_{1:T}, y_{1:T} \mid x_{1:T}) = p(\beta) p(\alpha_1) \prod_{t=2}^T p(\alpha_t \mid \alpha_{t-1}) \prod_{t=1}^T p(y_t \mid \alpha_t, \beta, \mathbf{x}_t).$$

(Equivalently, omit conditioning on $x_{1:T}$ if covariates are treated as fixed inputs.)

Part 5

1. (10%) Classify the following statements as True (T) or False (F).

- (a) In Bayesian inference, the posterior distribution is proportional to the likelihood times the prior.
- (b) The model evidence (marginal likelihood) $p(\mathcal{D})$ does not depend on the prior distribution.
- (c) If a prior is conjugate to the likelihood, then the posterior distribution belongs to the same parametric family as the prior.
- (d) Using a very strong prior makes the posterior less sensitive to the observed data.

Solution:

- (a) **T**. Bayes rule: $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta)$.
 - (b) **F**. $p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)p(\theta) d\theta$ depends on the prior.
 - (c) **T**. This is the definition of conjugacy.
 - (d) **T**. A strong prior dominates unless data are very informative.
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Mini-test 2

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Total points: 43

Please note that there are some questions with only one correct answer, whereas other questions require you to classify all given statements as either true or false.

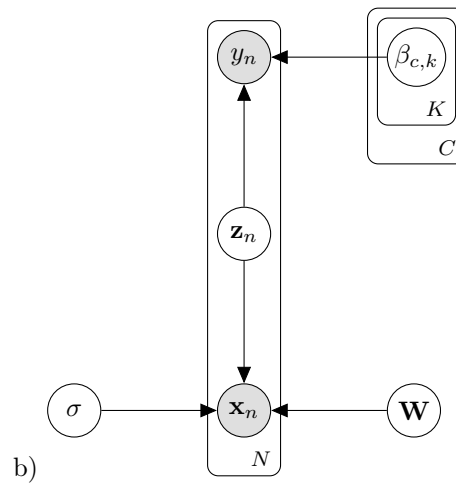
Part 1

PPCA is an unsupervised learning model. Suppose now that you would like to make it supervised - i.e., you would like to project each observation $\mathbf{x}_n \in \mathbb{R}^D$ to a lower dimensional projection $\mathbf{z}_n \in \mathbb{R}^K$, such that the projection $\mathbf{z}_n$ is also linked to a class label y_n . One way that you can do this, is by modifying the generative process of standard PPCA as follows (changes highlighted in blue and with $\leftarrow$ new!):

- Generative process:
 1. For each class $c \in \{\beta_1, \dots, \beta_C\} \leftarrow$ new!
 - (a) Sample weights $\beta_c \sim \mathcal{N}(\beta_c | \mathbf{0}, \tau \mathbf{I}) \leftarrow$ new!
 2. For each column $k \in \{1, \dots, K\}$ in the matrix $\mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_K)^T$
 - (a) Sample $\mathbf{w}_k \sim \mathcal{N}(\mathbf{w}_k | \mathbf{0}, \lambda \mathbf{I})$
 3. Sample $\sigma^2 \sim \text{InvGamma}(\sigma^2 | \alpha, \beta)$
 4. For each observation $n \in \{1, \dots, N\}$
 - (a) Sample latent projection $\mathbf{z}_n \sim \mathcal{N}(\mathbf{z}_n | \mathbf{0}, \mathbf{I})$
 - (b) Sample observation $\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | \mathbf{W} \mathbf{z}_n, \sigma^2 \mathbf{I})$
 - (c) Sample class label $y_n \sim \text{Categorical}(\text{Softmax}(\mathbf{z}_n), \beta_1, \dots, \beta_C) \leftarrow$ new!

where C denotes the number of classes.

1. [4 points] Which of the following PGMs is a correct representation of this new model? Only one option is correct.



Note that each β_c is a vector of length K , i.e. $\beta_c = \{\beta_{c,1}, \dots, \beta_{c,K}\}$. In the PGM, you can represent each $\beta_{c,k}$ and have the plate over K , or just go for representing β_c , in which case you wouldn't need the plate. But the two are equivalent.

2. [6 points] Consider the modified PPCA model from the previous question again. Classify the following statements as True (T) or False (F):

- (a) The modified PPCA model is not a generative model anymore because it is supervised. **F**
- (b) The relationship between latent projections $\mathbf{z}_n$ and class labels y_n is non-linear because of the Softmax. **F**
- (c) Given a dataset of N observations, $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^N$, we can compute the posterior distributions over all the latent variables in the model. **T**
- (d) With this modification, the learned latent projections $\mathbf{z}_n$ for each datapoint n will take into account both the observations $\mathbf{x}_n$ and the class label y_n . **T**
- (e) Supposed that you are given the posterior distribution over the latent projection $\mathbf{z}$. You can use this projection to predict the class label by computing $\arg \max_k z_k$. **F**
- (f) Once fitted, we can sample “fake observations” from this model by sampling $\mathbf{z} \sim \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I})$, and using $\mathbf{z}$ to obtain a “fake” $\mathbf{x}$ and its corresponding class label y . **T**

3. [6 points] Classify the following statements about generative models as True (T) or False (F):

- (a) Generative models are designed for dimensionality reduction tasks by preserving the most important features of the data. **F**
- (b) Generative models aim to learn the underlying probability distribution of the data and can be used to generate new samples similar to the training data. **T**
- (c) One of the advantages of generative models is their ability to handle missing data during training, allowing them to generate plausible samples even with incomplete information. **T**
- (d) Variational Autoencoders (VAEs) are a type of generative model that learns a latent representation of the input data, which can then be used to generate new samples. **T**
- (e) VAEs are typically trained using maximum likelihood estimation (MLE) to maximize the probability of observing the input data given the parameters of the model. **F**
- (f) Gaussian Mixture Models (GMMs) are a type of generative model that assumes the data is generated from a mixture of several Gaussian distributions. **T**
- (g) In VAEs, the decoder network maps input data to a latent space while the encoder network maps points in the latent space to the observation (data) space. **F**

Part 2

4. [6 points] Classify the following statements about MCMC as True (T) or False (F):

- (a) Importance sampling is a Monte Carlo method used for estimating properties of a target probability distribution by drawing samples from a different, but usually related, proposal distribution. **T**
- (b) A major advantage of importance sampling is that its efficiency does not depend on the choice of the proposal distribution. **F**
- (c) Metropolis-Hastings algorithm requires a proposal distribution from which to generate candidate samples, and it does not need to be symmetric. **T**
- (d) While the Metropolis-Hastings algorithm can converge to the target distribution under certain conditions, the choice of proposal distribution can significantly affect convergence speed and efficiency. **T**
- (e) Gibbs sampling always produces independent samples from the target distribution, ensuring convergence to the true distribution regardless of the number of iterations. **F**
- (f) Both Metropolis-Hastings and Gibbs sampling allow us to obtain a Gaussian approximation to the true posterior distribution. **F**

Part 3

5. [7 points] Classify the following statements about Gaussian processes (GPs) as True (T) or False (F):

- (a) GPs are parametric models that assume a specific functional form for the underlying relationship between input and output variables. **F**
- (b) GPs can be used to model target variables y that are count data by combining a Gaussian process prior with a Poisson likelihood. **T**
- (c) The marginal likelihood of a GP can be computed analytically even if the likelihood is non-Gaussian. **F**
- (d) One can find optimal hyper-parameters for the covariance function by maximizing the marginal likelihood of the model using a coordinate ascent algorithm. **T**
- (e) GPs are a family of non-parametric models used for regression and classification tasks, capable of capturing complex patterns in data without assuming a specific functional form for the underlying relationship. **T**
- (f) GPs provide point predictions without uncertainty estimates, making them unsuitable for tasks where understanding prediction confidence is crucial. **F**
- (g) GPs rely on a kernel function to compute the similarity between data points, with the choice of kernel influencing the flexibility and generalization ability of the model. **T**

Part 4

6. [4 points] Classify the following statements about reinforcement learning (RL) as True (T) or False (F):

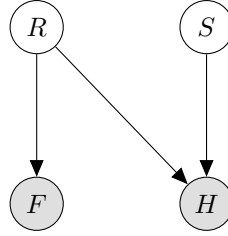
- (a) A Markov decision process describes a decision-making problem in which an agent interacts with an uncertain environment. Given the current state s and the selected action a , the next state s' is conditionally independent of all previous states and actions. **T**
- (b) A Markov decision process describes a decision-making problem in which an agent interacts with an uncertain environment. Given the current state s and the selected action a , the reward r depends also on all previous states and actions. **F**
- (c) The ϵ -greedy strategy in RL involves choosing the best action with probability ϵ . **F**
- (d) Model-free RL methods tend to require fewer interactions with the uncertain environment than model-based RL methods. **F**

Part 5

One morning, when Homer leaves his house, he realises that his grass is wet ($H = 1$). He knows that it can be either due to rain ($R = 1$), or because he has forgotten to turn off the sprinkler last night ($S = 1$).

Next he notices that the grass of his neighbour, Flanders, is also wet ($F = 1$).

The following PGM expresses Homer’s “domain knowledge” about the problem:



Let the prior probabilities for R and S be $P(R = 1) = 0.2$ and $P(S = 1) = 0.5$, and then the following conditional probability tables apply:

$P(F R)$		
	F=1	F=0
R=1	1	0
R=0	0.2	0.8

$P(H R, S)$		
	H=1	H=0
R=1, S=1	1	0
R=0, S=1	0.5	0.5
R=1, S=0	1	0
R=0, S=0	0	1

7. [6 points] Using Belief Propagation, calculate the probability that the grass is wet because Homer forgot to turn off the sprinkler. You can build on the following initial steps of the solution algorithm:

- $\lambda(F) = \pi(F) = \begin{cases} 0, & F = 0 \\ 1, & F = 1 \end{cases}$
- $\lambda(H) = \pi(H) = \begin{cases} 0, & H = 0 \\ 1, & H = 1 \end{cases}$
- $\pi(S) = \begin{cases} 0.5, & S = 0 \\ 0.5, & S = 1 \end{cases}$
- $\pi(R) = \begin{cases} 0.8, & R = 0 \\ 0.2, & R = 1 \end{cases}$
- $\lambda_{F,R}(R) = \sum_{F \in \{0,1\}} \lambda(F)P(F|R) = P(F = 1|R) = \begin{cases} 0.2, & R = 0 \\ 1, & R = 1 \end{cases}$
- $\pi_{R,H}(R) = \pi(R)\lambda_{F,R}(R) = \begin{cases} 0.16, & R = 0 \\ 0.2, & R = 1 \end{cases}$

Which of the following options corresponds to the correct solution for $\text{Bel}(S = 1) = P(S = 1|F = 1, H = 1)$? Only one option is correct.

- (a) $\frac{7}{12}$

Note that if your answer was $\frac{7}{50}$ then you forgot to normalize the beliefs at the end. Remember that $\text{Bel}(S = 1) \propto \lambda(S = 1)\pi(S = 1)$. It is a “proportional to” sign, not an equal sign.

Part 6

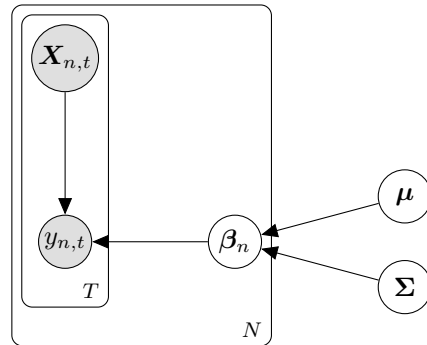


Figure 1: Model A

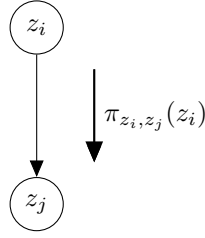
8. [4 points] Consider the model shown in Figure 1. Which of the following options corresponds to a variational distribution from the mean field family for the given model? Only one option is correct.

(a) $q(\boldsymbol{\mu})q(\boldsymbol{\Sigma}) \prod_{n=1}^N q(\beta_n)$

Note we ask for a mean-field approximation, so, in other words, a fully factorized distribution with no dependencies. Therefore, options with $q(\beta_n|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are wrong. Similarly, the option that included $q(y_{n,t}|\beta_n, X_{n,t})$ was incorrect because $y_{n,t}$ is an observed variable, and not a latent variable which we wish to infer.

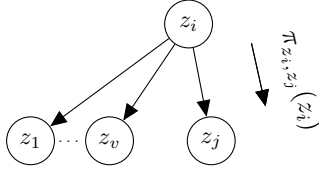
π messages - from parent to child

Simple version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i)$$

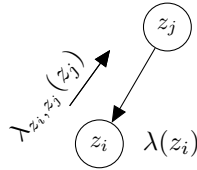
Complete version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i) \prod_{v \neq j} \lambda_{z_v, z_i}(z_i)$$

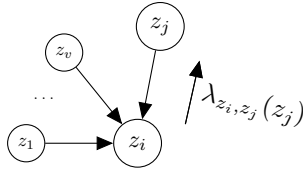
λ messages - from child to parent

Simple version:



$$\lambda_{z_i, z_j}(z_j) = \sum_{z_i} \lambda(z_i) p(z_i | z_j)$$

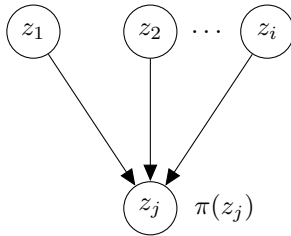
Complete version:



$$\lambda_{z_i, z_j}(z_j) = \sum_{z_i} \lambda(z_i) \sum_{z_v \in \mathbf{z} \setminus z_j} p(z_i | z_1, z_2, \dots, z_j) \prod_{v \neq j} \pi_{z_v, z_i}(z_v)$$

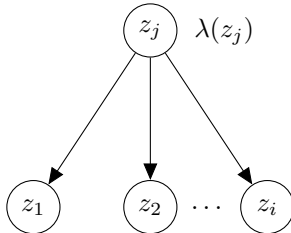
Belief components

Causal:



$$\pi(z_j) = \sum_{z_i \in \text{parent}(z_j)} p(z_j | z_1, z_2, \dots) \prod_{z_i \in \text{parent}(z_j)} \pi_{z_i, z_j}(z_i)$$

Diagnostic:



$$\lambda(z_j) = \prod_{z_i \in \text{child}(z_j)} \lambda_{z_i, z_j}(z_j)$$

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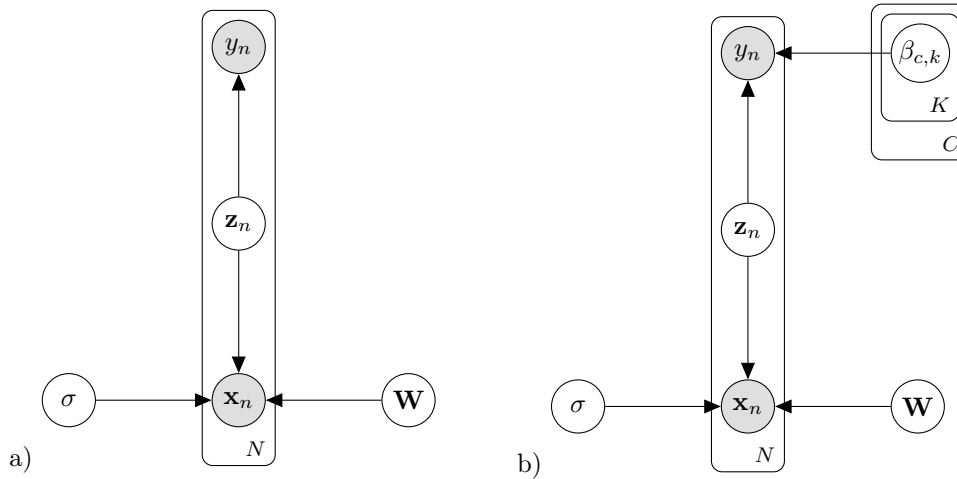
Part 1

PPCA is an unsupervised learning model. Suppose now that you would like to make it supervised - i.e., you would like to project each observation $\mathbf{x}_n \in \mathbb{R}^D$ to a lower dimensional projection $\mathbf{z}_n \in \mathbb{R}^K$, such that the projection $\mathbf{z}_n$ is also linked to a class label y_n . One way that you can do this, is by modifying the generative process of standard PPCA as follows (changes highlighted in blue and with $\leftarrow$ new!):

- Generative process:
 1. For each class $c \in \{\beta_1, \dots, \beta_C\} \leftarrow$ new!
 - (a) Sample weights $\beta_c \sim \mathcal{N}(\beta_c | \mathbf{0}, \tau \mathbf{I}) \leftarrow$ new!
 2. For each column $k \in \{1, \dots, K\}$ in the matrix $\mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_K)^T$
 - (a) Sample $\mathbf{w}_k \sim \mathcal{N}(\mathbf{w}_k | \mathbf{0}, \lambda \mathbf{I})$
 3. Sample $\sigma^2 \sim \text{InvGamma}(\sigma^2 | \alpha, \beta)$
 4. For each observation $n \in \{1, \dots, N\}$
 - (a) Sample latent projection $\mathbf{z}_n \sim \mathcal{N}(\mathbf{z}_n | \mathbf{0}, \mathbf{I})$
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where C denotes the number of classes.

1. [4 points] Which of the following PGMs is a correct representation of this new model? Only one option is correct.





- (a) The modified PPCA model is not a generative model anymore because it is supervised.
- (b) The relationship between latent projections $\mathbf{z}_n$ and class labels y_n is non-linear because of the Softmax.
- (c) Given a dataset of N observations, $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^N$, we can compute the posterior distributions over all the latent variables in the model.
- (d) With this modification, the learned latent projections $\mathbf{z}_n$ for each datapoint n will take into account both the observations $\mathbf{x}_n$ and the class label y_n .
- (e) Supposed that you are given the posterior distribution over the latent projection $\mathbf{z}$. You can use this projection to predict the class label by computing $\arg \max_k z_k$.
- (f) Once fitted, we can sample “fake observations” from this model by sampling $\mathbf{z} \sim \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I})$, and using $\mathbf{z}$ to obtain a “fake” $\mathbf{x}$ and its corresponding class label y .

- (a) Generative models are designed for dimensionality reduction tasks by preserving the most important features of the data.
- (b) Generative models aim to learn the underlying probability distribution of the data and can be used to generate new samples similar to the training data.
- (c) One of the advantages of generative models is their ability to handle missing data during training, allowing them to generate plausible samples even with incomplete information.
- (d) Variational Autoencoders (VAEs) are a type of generative model that learns a latent representation of the input data, which can then be used to generate new samples.
- (e) VAEs are typically trained using maximum likelihood estimation (MLE) to maximize the probability of observing the input data given the parameters of the model.
- (f) Gaussian Mixture Models (GMMs) are a type of generative model that assumes the data is generated from a mixture of several Gaussian distributions.
- (g) In VAEs, the decoder network maps input data to a latent space while the encoder network maps points in the latent space to the observation (data) space.

Part 2

4. [6 points] Classify the following statements about MCMC as True (T) or False (F):

- (a) Importance sampling is a Monte Carlo method used for estimating properties of a target probability distribution by drawing samples from a different, but usually related, proposal distribution.
- (b) A major advantage of importance sampling is that its efficiency does not depend on the choice of the proposal distribution.
- (c) Metropolis-Hastings algorithm requires a proposal distribution from which to generate candidate samples, and it does not need to be symmetric.
- (d) While the Metropolis-Hastings algorithm can converge to the target distribution under certain conditions, the choice of proposal distribution can significantly affect convergence speed and efficiency.
- (e) Gibbs sampling always produces independent samples from the target distribution, ensuring convergence to the true distribution regardless of the number of iterations.
- (f) Both Metropolis-Hastings and Gibbs sampling allow us to obtain a Gaussian approximation to the true posterior distribution.

Part 3

5. [7 points] Classify the following statements about Gaussian processes (GPs) as True (T) or False (F):

- (a) GPs are parametric models that assume a specific functional form for the underlying relationship between input and output variables.
- (b) GPs can be used to model target variables y that are count data by combining a Gaussian process prior with a Poisson likelihood.
- (c) The marginal likelihood of a GP can be computed analytically even if the likelihood is non-Gaussian.
- (d) One can find optimal hyper-parameters for the covariance function by maximizing the marginal likelihood of the model using a coordinate ascent algorithm.
- (e) GPs are a family of non-parametric models used for regression and classification tasks, capable of capturing complex patterns in data without assuming a specific functional form for the underlying relationship.
- (f) GPs provide point predictions without uncertainty estimates, making them unsuitable for tasks where understanding prediction confidence is crucial.
- (g) GPs rely on a kernel function to compute the similarity between data points, with the choice of kernel influencing the flexibility and generalization ability of the model.

Part 4

6. [4 points] Classify the following statements about reinforcement learning (RL) as True (T) or False (F):

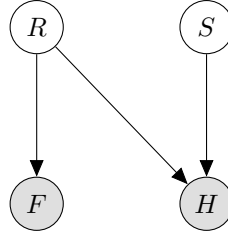
- (a) A Markov decision process describes a decision-making problem in which an agent interacts with an uncertain environment. Given the current state s and the selected action a , the next state s' is conditionally independent of all previous states and actions.
- (b) A Markov decision process describes a decision-making problem in which an agent interacts with an uncertain environment. Given the current state s and the selected action a , the reward r depends also on all previous states and actions.
- (c) The ϵ -greedy strategy in RL involves choosing the best action with probability ϵ .
- (d) Model-free RL methods tend to require fewer interactions with the uncertain environment than model-based RL methods.

Part 5

One morning, when Homer leaves his house, he realises that his grass is wet ($H = 1$). He knows that it can be either due to rain ($R = 1$), or because he has forgotten to turn off the sprinkler last night ($S = 1$).

Next he notices that the grass of his neighbour, Flanders, is also wet ($F = 1$).

The following PGM expresses Homer’s “domain knowledge” about the problem:



Let the prior probabilities for R and S be $P(R = 1) = 0.2$ and $P(S = 1) = 0.5$, and then the following conditional probability tables apply:

$P(F R)$		
	F=1	F=0
R=1	1	0
R=0	0.2	0.8

$P(H R, S)$		
	H=1	H=0
R=1, S=1	1	0
R=0, S=1	0.5	0.5
R=1, S=0	1	0
R=0, S=0	0	1

7. [6 points] Using Belief Propagation, calculate the probability that the grass is wet because Homer forgot to turn off the sprinkler. You can build on the following initial steps of the solution algorithm:

- $\lambda(F) = \pi(F) = \begin{cases} 0, & F = 0 \\ 1, & F = 1 \end{cases}$
- $\lambda(H) = \pi(H) = \begin{cases} 0, & H = 0 \\ 1, & H = 1 \end{cases}$
- $\pi(S) = \begin{cases} 0.5, & S = 0 \\ 0.5, & S = 1 \end{cases}$
- $\pi(R) = \begin{cases} 0.8, & R = 0 \\ 0.2, & R = 1 \end{cases}$
- $\lambda_{F,R}(R) = \sum_{F \in \{0,1\}} \lambda(F)P(F|R) = P(F = 1|R) = \begin{cases} 0.2, & R = 0 \\ 1, & R = 1 \end{cases}$
- $\pi_{R,H}(R) = \pi(R)\lambda_{F,R}(R) = \begin{cases} 0.16, & R = 0 \\ 0.2, & R = 1 \end{cases}$

Which of the following options corresponds to the correct solution for $\text{Bel}(S = 1) = P(S = 1|F = 1, H = 1)$? Only one option is correct.

- (a) $\frac{1}{5}$
- (b) $\frac{7}{25}$
- (c) $\frac{1}{10}$
- (d) $\frac{7}{50}$
- (e) $\frac{5}{12}$
- (f) $\frac{7}{12}$

Part 6

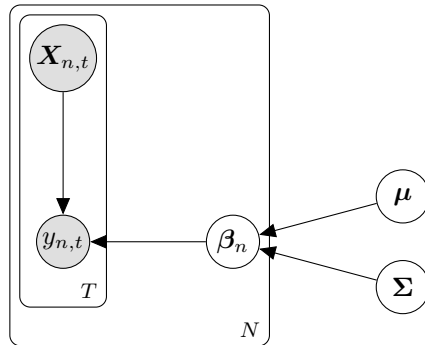


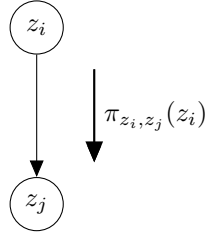
Figure 1: Model A

8. [4 points] Consider the model shown in Figure 1. Which of the following options corresponds to a variational distribution from the mean field family for the given model? Only one option is correct.

- (a) $q(\boldsymbol{\mu})q(\boldsymbol{\Sigma}) \prod_{n=1}^N q(\beta_n | \boldsymbol{\mu}, \boldsymbol{\Sigma})$
- (b) $q(\boldsymbol{\mu})q(\boldsymbol{\Sigma}) \left(\prod_{n=1}^N q(\beta_n | \boldsymbol{\mu}, \boldsymbol{\Sigma}) \prod_{t=1}^T q(y_{n,t} | \beta_n, X_{n,t}) \right)$
- (c) $q(\boldsymbol{\mu})q(\boldsymbol{\Sigma}) \prod_{n=1}^N q(\beta_n)$
- (d) $q(\boldsymbol{\mu})q(\boldsymbol{\Sigma})q(\beta_n)$

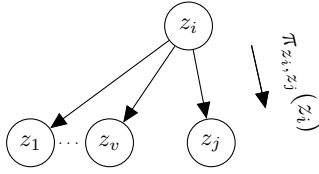
π messages - from parent to child

Simple version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i)$$

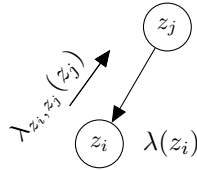
Complete version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i) \prod_{v \neq j} \lambda_{z_v, z_i}(z_i)$$

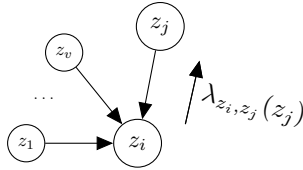
λ messages - from child to parent

Simple version:



$$\lambda_{z_i, z_j}(z_j) = \sum_{z_i} \lambda(z_i) p(z_i | z_j)$$

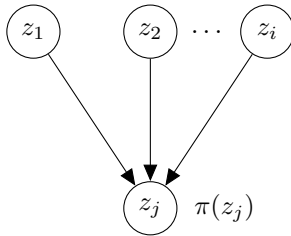
Complete version:



$$\lambda_{z_i, z_j}(z_j) = \sum_{z_i} \lambda(z_i) \sum_{z_v \in \mathbf{z} \setminus z_j} p(z_i | z_1, z_2, \dots, z_j) \prod_{v \neq j} \pi_{z_v, z_i}(z_v)$$

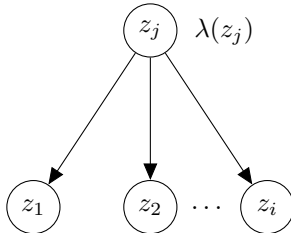
Belief components

Causal:



$$\pi(z_j) = \sum_{z_i \in \text{parent}(z_j)} p(z_j | z_1, z_2, \dots) \prod_{z_i \in \text{parent}(z_j)} \pi_{z_i, z_j}(z_i)$$

Diagnostic:



$$\lambda(z_j) = \prod_{z_i \in \text{child}(z_j)} \lambda_{z_i, z_j}(z_j)$$

Technical University of Denmark

SPRING, 2025
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MODEL-BASED MACHINE LEARNING

Mini-test 2

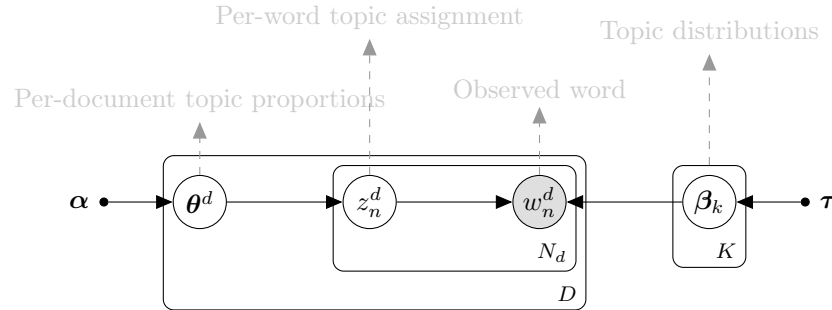
(Time allowed: 60 MINUTES)

Name: _____

Student ID: _____

Part 1 - Modelling

Consider the following PGM and generative process corresponding to the Latent Dirichlet Allocation (LDA) model that you studied in class (notation may differ slightly):



Generative process of LDA model:

1. For each topic k
 - (a) Draw topic's distribution over words $\beta_k \sim \text{Dirichlet}(\beta_k | \tau)$
2. For each document $d \in \{1, \dots, D\}$
 - (a) Draw topic proportions $\theta^d \sim \text{Dirichlet}(\theta^d | \alpha)$
 - (b) For the n^{th} word
 - i. Draw topic assignment $z_n^d \sim \text{Categorical}(z_n^d | \theta^d)$
 - ii. Draw word $w_n^d \sim \text{Categorical}(w_n^d | \beta_{z_n^d})$

1. (10%) As you know, LDA assigns a distribution over topics $\theta^d \in \mathbb{R}^K$ to each document d . However, we typically want each document to be dominated by only a small subset of the K topics, as this makes the results more interpretable and also aligns with our domain knowledge. Having this in mind, how should you set the hyper-parameter α ? Justify.

R: We want to encourage sparsity in the vector θ^d . A Dirichlet prior with $\alpha < 1$ achieves this by placing most of the probability density on the corners of the simplex.

2. (20%) Suppose now that each document corresponds to a news article on a website and that, besides the document text $\mathbf{w}^d$, you also have access to the number of clicks $c^d \in \mathbb{N}_0$ in the article's link.

- You would like the learned topics also to be good predictors of the number of clicks (and, in that way, discover topics that make news articles popular), so you decide to extend the LDA model to include a linear parametric relationship between $\boldsymbol{\theta}^d$ and c^d . In this way, the topic proportions $\boldsymbol{\theta}^d$ of each document will have to explain both the observed words in that document and its clicks c^d (since both c^d and $\mathbf{w}^d$ will be influenced by $\boldsymbol{\theta}^d$). Choose an appropriate distribution for c^d .
- Suppose that you also have access to a vector $\mathbf{x}^d$ containing information about what time of day the news article was published on the website, where in the website it was listed (e.g., highlighted news piece or not), etc. You expect this input information to further shift the number of clicks of a news article up or down, but in a non-linear way, so you decide to model the relationship between $\mathbf{x}^d$ and c^d with a Bayesian neural network with parameters ϕ .

Write the generative process of the revised model that incorporates these two extensions. Make sure to choose appropriate distributions. Hint: it can be helpful to quickly sketch out the new PGM to understand this new (revised) model.

R: A few things to note from the question's text:

- $c^d \in \mathbb{N}_0$, so we need to choose a probability distribution that is appropriate for count data - e.g., Poisson or Negative-Binomial
- If you chosen Poisson, note that the parameter of the Poisson (rate) must be positive, so we need to choose a link-function that ensures that (e.g., a log-link function)
- For the linear parametric dependency of c^d on $\boldsymbol{\theta}^d$ we need to introduce a new vector of parameters (let's call them $\boldsymbol{\eta}$) and a reasonable prior for them (e.g. Normal or Cauchy)
- For the non-linear parametric dependency of c^d on $\mathbf{x}^d$ we need to introduce a new set of parameters (let's call them ϕ) and a reasonable prior for them (e.g. Normal or Cauchy)

Here is a possible solution:

- Draw vector of parameters for linear function $\boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{\eta}|\mathbf{0}, \sigma\mathbf{I})$
- Draw neural network parameters $\phi \sim \mathcal{N}(\phi|\mathbf{0}, \sigma\mathbf{I})$
- For each topic k
 - Draw topic's distribution over words $\beta_k \sim \text{Dirichlet}(\beta_k|\boldsymbol{\tau})$
- For each document $d \in \{1, \dots, D\}$
 - Draw topic proportions $\boldsymbol{\theta}^d \sim \text{Dirichlet}(\boldsymbol{\theta}^d|\boldsymbol{\alpha})$
 - For the n^{th} word
 - Draw topic assignment $z_n^d \sim \text{Multinomial}(z_n^d|\boldsymbol{\theta}^d)$
 - Draw word $w_n^d \sim \text{Multinomial}(w_n^d|\beta_{z_n^d})$
 - Draw number of clicks $c^d \sim \text{Poisson}(c^d|\exp(\phi^T \boldsymbol{\theta}^d + \text{NNet}_\phi(\mathbf{x}^d)))$

3. (20%) Classify the following statements related to Stochastic Processes as True (T) or False (F):

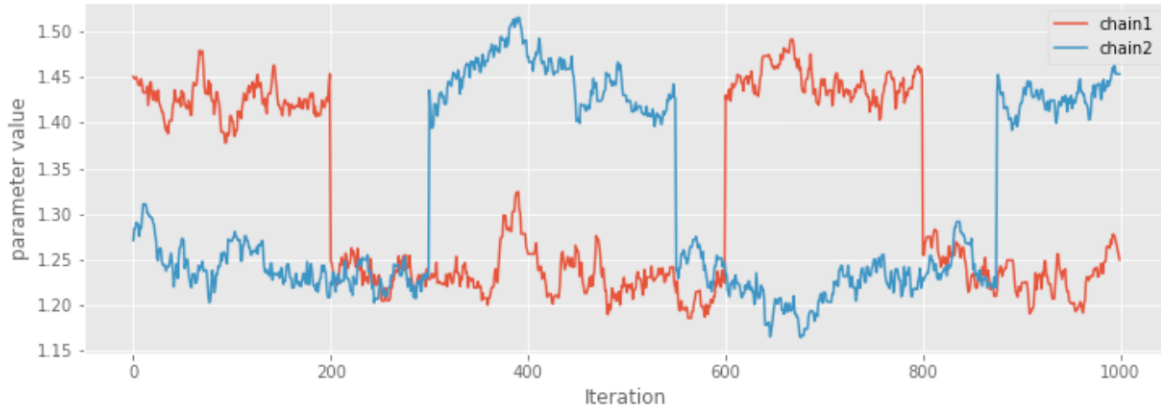
- A) A Dirichlet Process is a distribution over distributions, often used in Bayesian nonparametrics. T
- B) In the stick-breaking construction of a Dirichlet Process, the weights (“pieces of the stick”) are generated using Beta-distributed random variables. T
- C) Samples drawn from a Dirichlet Process are always continuous probability distributions. F
- D) A Gaussian Process defines a distribution over functions, where any finite set of function values has a joint Gaussian distribution. T
- E) The posterior mean of a Gaussian Process regression is always a linear function of the input data. F
- F) Gaussian Processes can naturally quantify uncertainty in predictions. T
- G) A Gaussian Process assumes that outputs are conditionally independent given the inputs. F
- H) A Gaussian Process prior always leads to a Gaussian posterior, regardless of the likelihood function. F

Part 2 - Approximate inference

1. (15%) Classify the following statements related to Monte Carlo methods for Bayesian inference as True (T) or False (F):

- A) Monte Carlo methods allow us to approximate expectations of some function $f(\mathbf{z})$ under a probability distribution $p(\mathbf{z}|\mathbf{x})$ as finite sums, where $\mathbf{z}$ denotes the latent variables in the model and $\mathbf{x}$ the observations. T
- B) Importance sampling does NOT allow for drawing samples from the posterior distribution $p(\mathbf{z}|\mathbf{x})$, but it allows for approximating expectations with respect to $p(\mathbf{z}|\mathbf{x})$. T
- C) The acceptance probability in Metropolis-Hastings depends only on the likelihood of the new proposed state, not on the current state. F
- D) In Hamiltonian Monte Carlo, momentum variables are introduced to help explore the parameter space more efficiently, leading to samples with lower autocorrelation. T
- E) Gibbs sampling is an MCMC algorithm with no rejection because it satisfies detailed balance. F
- F) When using Gibbs sampling for inference, it is necessary that you use conjugate priors in the model. F

2. (10%) While using MCMC to perform approximate Bayesian inference on another model, you decided to use 2 chains in parallel (e.g. to take advantage of more CPU cores). However, when inspecting the traceplots, you obtained the following plot.

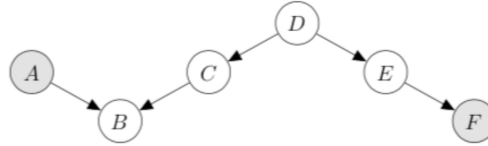


Which of the options below can you conclude?

- A) The posterior distribution is highly skewed
 - B) The posterior distribution is likely multimodal - CORRECT
 - C) The posterior distribution is intractable
 - D) There is a problem with the initialization of the MCMC algorithm
 - E) You should use thinning
3. (10%) Classify the following statements related to Variational Inference (VI) as True (T) or False (F):
- A) In a model with a multimodal posterior distribution (e.g., a Gaussian Mixture Model), we are always interested in capturing all the posterior modes. Therefore, performing inference with VI will always be problematic. F
 - B) For VI, using conjugate priors in a model is important when deriving analytical expressions for the coordinated ascent updates for optimizing the ELBO. T
 - C) Minimizing the $\mathbb{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x}))$ is equivalent to minimizing the ELBO. F
 - D) The choice of the variational family affects the accuracy of the approximation to the true posterior. T

Part 3 - Exact inference

Consider the following PGM:



All variables are binary, except for D, which can take 3 values. The conditional probability tables (CPTs) for the different variables are given below:

$d = 0$	$d = 1$	$d = 2$
0.3	0.6	0.1

Table 1: $P(D = d)$

	$c = 0$	$c = 1$
d=0	0.3	0.7
d=1	0.8	0.2
d=2	0.4	0.6

Table 2: $P(C = c|D = d)$

	$e = 0$	$e = 1$
d=0	0.5	0.5
d=1	0.1	0.9
d=2	0.7	0.3

Table 4: $P(E = e|D = d)$

	$b = 0$	$b = 1$
a=0, c=0	0.1	0.9
a=0, c=1	0.5	0.5
a=1, c=0	0.4	0.6
a=1, c=1	0.3	0.7

Table 3: $P(B = b|A = a, C = c)$

	$f = 0$	$f = 1$
e=0	0.6	0.4
e=1	0.8	0.2

Table 5: $P(F = f|E = e)$

- (15%) You observed the values $A = 0$ and $F = 1$, and now you would like to perform exact inference on the remaining latent variables using Belief Propagation (BP). In order to make things easier, we have pre-calculated some quantities:

$$\pi(c = 0) = 0.5 \quad \pi(c = 1) = 0.1$$

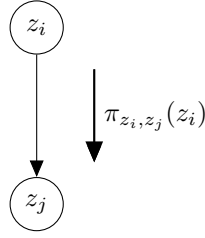
$$\lambda_{C,D}(D) = 1$$

What should the table for $\pi(B)$ look like?

- $\pi(B = 0) = 0.09$ $\pi(B = 1) = 0.49$
- $\pi(B = 0) = 0.10$ $\pi(B = 1) = 0.50$ - CORRECT
- $\pi(B = 0) = 0.50$ $\pi(B = 1) = 0.10$
- $\pi(B = 0) = 0.09$ $\pi(B = 1) = 0.50$
- $\pi(B = 0) = 0.10$ $\pi(B = 1) = 0.49$

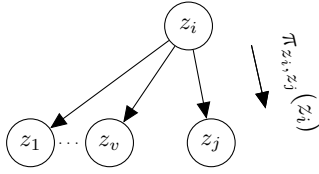
π messages - from parent to child

Simple version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i)$$

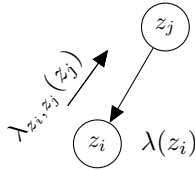
Complete version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i) \prod_{v \neq j} \lambda_{z_v, z_i}(z_i)$$

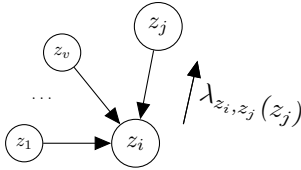
λ messages - from child to parent

Simple version:



$$\lambda_{z_i, z_j}(z_j) = \sum_{z_i} \lambda(z_i) p(z_i | z_j)$$

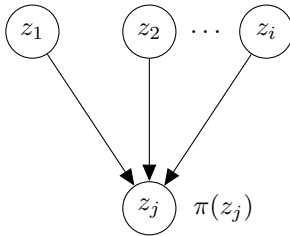
Complete version:



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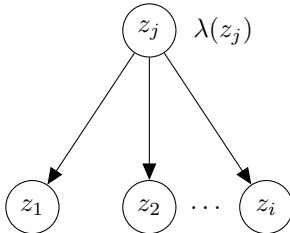
Belief components

Causal:



$$\pi(z_j) = \sum_{z_i \in \text{parent}(z_j)} p(z_j | z_1, z_2, \dots) \prod_{z_i \in \text{parent}(z_j)} \pi_{z_i, z_j}(z_i)$$

Diagnostic:



$$\lambda(z_j) = \prod_{z_i \in \text{child}(z_j)} \lambda_{z_i, z_j}(z_j)$$

Technical University of Denmark

SPRING, 2025
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MODEL-BASED MACHINE LEARNING

Mini-test 2

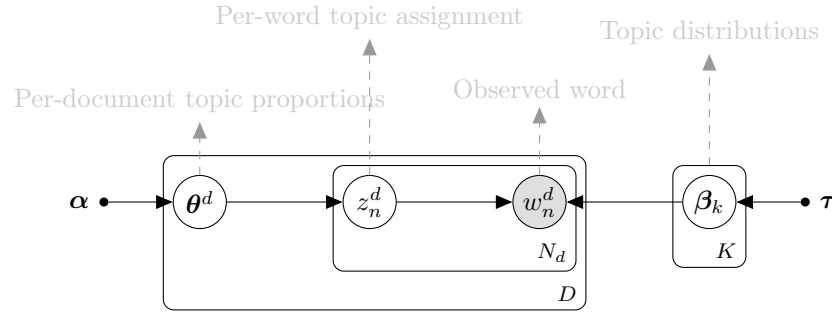
(Time allowed: 60 MINUTES)

Name: _____

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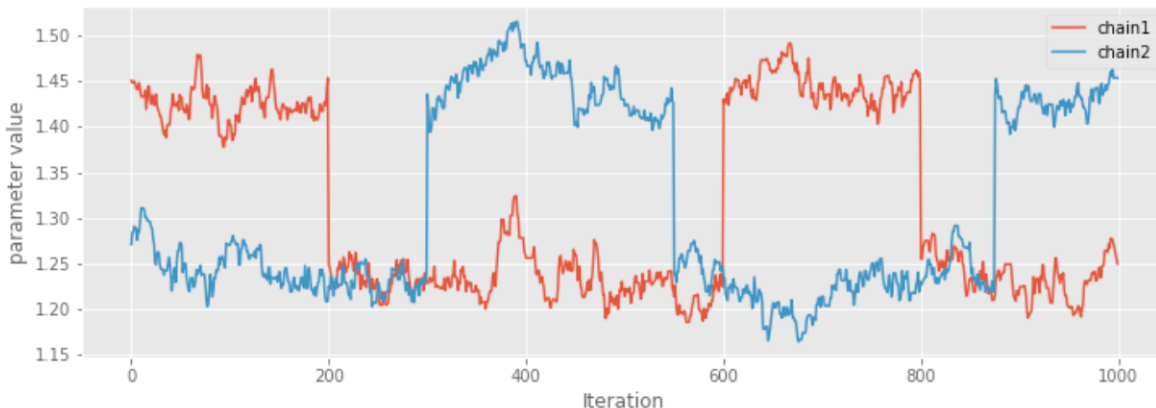
Write the generative process of the revised model that incorporates these two extensions. Make sure to choose appropriate distributions. Hint: it can be helpful to quickly sketch out the new PGM to understand this new (revised) model.

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Part 2 - Approximate inference

1. (15%) Classify the following statements related to Monte Carlo methods for Bayesian inference as True (T) or False (F):
 - A) Monte Carlo methods allow us to approximate expectations of some function $f(\mathbf{z})$ under a probability distribution $p(\mathbf{z}|\mathbf{x})$ as finite sums, where $\mathbf{z}$ denotes the latent variables in the model and $\mathbf{x}$ the observations.
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 - F) When using Gibbs sampling for inference, it is necessary that you use conjugate priors in the model.
2. (10%) While using MCMC to perform approximate Bayesian inference on another model, you decided to use 2 chains in parallel (e.g. to take advantage of more CPU cores). However, when inspecting the traceplots, you obtained the following plot.

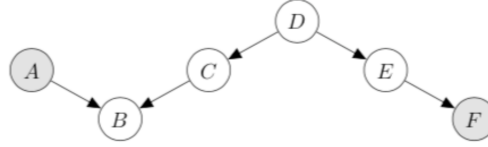


Which of the options below can you conclude?

- A) The posterior distribution is highly skewed
 - B) The posterior distribution is likely multimodal
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 - D) There is a problem with the initialization of the MCMC algorithm
 - E) You should use thinning
3. (10%) Classify the following statements related to Variational Inference (VI) as True (T) or False (F):
 - A) In a model with a multimodal posterior distribution (e.g., a Gaussian Mixture Model), we are always interested in capturing all the posterior modes. Therefore, performing inference with VI will always be problematic.
 - B) For VI, using conjugate priors in a model is important when deriving analytical expressions for the coordinated ascent updates for optimizing the ELBO.
 - C) Minimizing the $\mathbb{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x}))$ is equivalent to minimizing the ELBO.
 - D) The choice of the variational family affects the accuracy of the approximation to the true posterior.

Part 3 - Exact inference

Consider the following PGM:



All variables are binary, except for D, which can take 3 values. The conditional probability tables (CPTs) for the different variables are given below:

$d = 0$	$d = 1$	$d = 2$
0.3	0.6	0.1

Table 1: $P(D = d)$

	$c = 0$	$c = 1$
d=0	0.3	0.7
d=1	0.8	0.2
d=2	0.4	0.6

Table 2: $P(C = c|D = d)$

	$e = 0$	$e = 1$
d=0	0.5	0.5
d=1	0.1	0.9
d=2	0.7	0.3

Table 4: $P(E = e|D = d)$

	$b = 0$	$b = 1$
a=0, c=0	0.1	0.9
a=0, c=1	0.5	0.5
a=1, c=0	0.4	0.6
a=1, c=1	0.3	0.7

Table 3: $P(B = b|A = a, C = c)$

	$f = 0$	$f = 1$
e=0	0.6	0.4
e=1	0.8	0.2

Table 5: $P(F = f|E = e)$

1. (15%) You observed the values $A = 0$ and $F = 1$, and now you would like to perform exact inference on the remaining latent variables using Belief Propagation (BP). In order to make things easier, we have pre-calculated some quantities:

$$\pi(c = 0) = 0.5 \quad \pi(c = 1) = 0.1$$

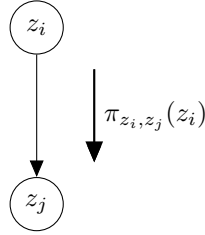
$$\lambda_{C,D}(D) = 1$$

What should the table for $\pi(B)$ look like?

- A) $\pi(B = 0) = 0.09$ $\pi(B = 1) = 0.49$
 B) $\pi(B = 0) = 0.10$ $\pi(B = 1) = 0.50$
 C) $\pi(B = 0) = 0.50$ $\pi(B = 1) = 0.10$
 D) $\pi(B = 0) = 0.09$ $\pi(B = 1) = 0.50$
 E) $\pi(B = 0) = 0.10$ $\pi(B = 1) = 0.49$

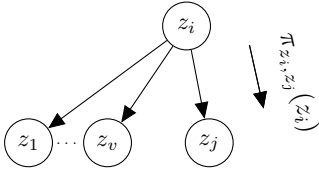
π messages - from parent to child

Simple version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i)$$

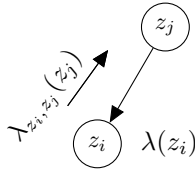
Complete version:



$$\pi_{z_i, z_j}(z_i) = \pi(z_i) \prod_{v \neq j} \lambda_{z_v, z_i}(z_i)$$

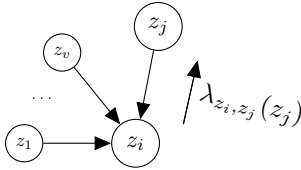
λ messages - from child to parent

Simple version:



$$\lambda_{z_i, z_j}(z_j) = \sum_{z_i} \lambda(z_i) p(z_i | z_j)$$

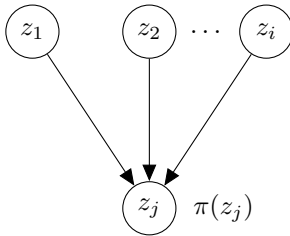
Complete version:



$$\lambda_{z_i, z_j}(z_j) = \sum_{z_i} \lambda(z_i) \sum_{z_v \in \mathbf{z} \setminus z_j} p(z_i | z_1, z_2, \dots, z_j) \prod_{v \neq j} \pi_{z_v, z_i}(z_v)$$

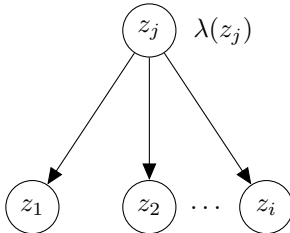
Belief components

Causal:



$$\pi(z_j) = \sum_{z_i \in \text{parent}(z_j)} p(z_j | z_1, z_2, \dots) \prod_{z_i \in \text{parent}(z_j)} \pi_{z_i, z_j}(z_i)$$

Diagnostic:



$$\lambda(z_j) = \prod_{z_i \in \text{child}(z_j)} \lambda_{z_i, z_j}(z_j)$$

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MODEL-BASED MACHINE LEARNING

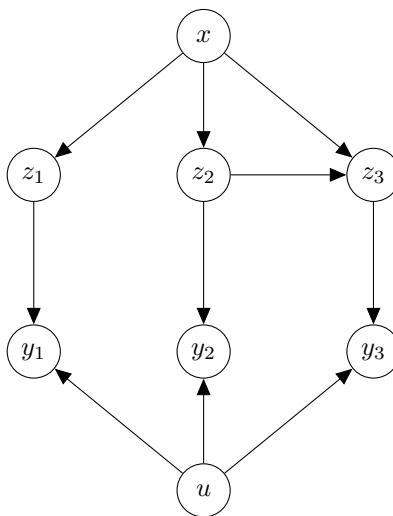
Mini-test 1

(Time allowed: 60 MINUTES)

Name: _____
Student ID: _____

Part 1

Consider the following probabilistic graphical model (PGM):



1. (10%) Write the factorization of the joint distribution according to the PGM described above.

$$p(x)p(u)p(z_1|x)p(z_2|x)p(z_3|z_2,x)p(y_1|z_1,u)p(y_2|z_2,u)p(y_3|z_3,u)$$

2. (20%) Which of the following expressions correspond to valid expressions for $p(z_3|z_2, u)$ (more than option may be correct):

(a) $\frac{\sum_x p(x)p(z_2|x)p(z_3|z_2,x)}{\sum_x p(x)p(z_2|x)}$

3. (15%) Classify the following statement as “True”, “False”, or “Cannot prove”:

- (a) $u \perp\!\!\!\perp z_3 | \{y_3, x\}$ - cannot prove
- (b) $x \perp\!\!\!\perp y_1 | \{z_1, y_2, y_3\}$ - cannot prove

Part 2

Consider the following generative process that models observed count data $\{y_{i,n}\}_{n=1}^{N_i}$ for each individual i using a hierarchical modelling approach:

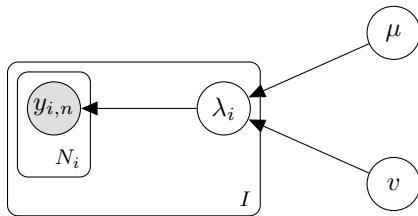
1. Draw $\mu \sim \mathcal{N}(0, 1)$
2. Draw $v \sim \text{Inverse-Gamma}(1, 1)$
3. For each individual $i \in \{1, \dots, I\}$:
 - (a) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mu, v)$
 - (b) For each observation $n \in \{1, \dots, N_i\}$ of the individual i :
 - i. Draw observed count $y_{i,n} \sim \text{Poisson}(\exp(\lambda_i))$

Make sure that you understand it well.

1. (15%) Write the factorization of the joint distribution according to the model described above.

$$p(\mu)p(v) \prod_{i=1}^I p(\lambda_i | \mu, v) \prod_{n=1}^{N_i} p(y_{i,n} | \lambda_i)$$

2. (15%) Draw the PGM representation corresponding to the generative process described above.



3. (10%) Considering the generative process of the model presented above, classify the following statements as True (T) or False (F).

(a) $p(\mu, \boldsymbol{\lambda}) = \int_v \int_{\mathbf{y}} p(\mu, v, \boldsymbol{\lambda}, \mathbf{y})$ - F

(b) $p(\boldsymbol{\lambda} | \mathbf{y}, \mu, v) \propto p(\mu, v, \boldsymbol{\lambda}, \mathbf{y})$ - T

(c) $p(\mu | \mathbf{y}, \boldsymbol{\lambda}, v) \propto p(\mu, v, \boldsymbol{\lambda}, \mathbf{y})$ - T

(d) $p(\mu | \boldsymbol{\lambda}) = \frac{\int_v p(\mu)p(v)p(\boldsymbol{\lambda}|\mu,v)}{\int_v \int_{\boldsymbol{\lambda}} p(\mu)p(v)p(\boldsymbol{\lambda}|\mu,v)}$ - F

4. (15%) Suppose now that you also have access to covariates $\mathbf{x}_i$ that describe characteristics of the each individual i that could help explain his/her latent log-rate λ_i . You may assume that the relationship between the mean of λ_i and $\mathbf{x}_i$ is linear in parameters $\boldsymbol{\beta} = \{\beta_1, \beta_2, \dots, \beta_D\}$. Which of the following generative processes corresponds to this enhanced model?

Option X:

(a) Draw $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma \mathbf{I})$

(b) Draw $v \sim \text{Inverse-Gamma}(1, 1)$

(c) For each individual $i \in \{1, \dots, I\}$:

(i) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\boldsymbol{\beta}^T \mathbf{x}_i, v)$

(ii) For each observation $n \in \{1, \dots, N_i\}$ of the individual i :

A. Draw observed count $y_{i,n} \sim \text{Poisson}(\exp(\lambda_i))$

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MODEL-BASED MACHINE LEARNING

Mini-test 1

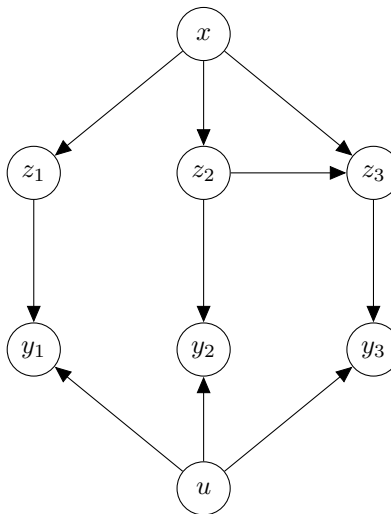
(Time allowed: 60 MINUTES)

Name: _____

Student ID: _____

Part 1

Consider the following probabilistic graphical model (PGM):



1. (10%) Write the factorization of the joint distribution according to the PGM described above.

2. (20%) Which of the following expressions correspond to valid expressions for $p(z_3|z_2, u)$ (more than option may be correct):

- (a) $\frac{\sum_x \sum_{z_1} \sum_{y_1} \sum_{y_2} p(y, x, z_1, z_2, z_3, y_1, y_2, y_3)}{\sum_x \sum_{z_1} \sum_{z_3} \sum_{y_1} \sum_{y_2} p(y, x, z_1, z_2, z_3, y_1, y_2, y_3)}$
- (b) $\sum_{y_1} p(z_3|z_2, u, y_1)$
- (c) $\frac{\sum_x p(x)p(z_2|x)p(z_3|z_2, x)}{\sum_x p(x)p(z_2|x)}$
- (d) $\sum_{y_2} p(z_2|x)p(z_3|z_2, x)p(y_2|z_2, u)$

3. (15%) Classify the following statement as “True”, “False”, or “Cannot prove”:

(a) $u \perp\!\!\!\perp z_3 \mid \{y_3, x\}$

(b) $x \perp\!\!\!\perp y_1 \mid \{z_1, y_2, y_3\}$

Consider the following generative process that models observed count data $\{y_{i,n}\}_{n=1}^{N_i}$ for each individual i using a hierarchical modelling approach:

- Make sure that you understand it well.

1. (15%) Write the factorization of the joint distribution according to the model described above.
2. (15%) Draw the PGM representation corresponding to the generative process described above.

3. (10%) Considering the generative process of the model presented above, classify the following statements as True (T) or False (F).

- (a) $p(u, \boldsymbol{\lambda}) = \int_v \int_{\mathbf{y}} p(u, v, \boldsymbol{\lambda}, \mathbf{y})$
- (b) $p(\boldsymbol{\lambda} | \mathbf{y}, u, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$
- (c) $p(u | \mathbf{y}, \boldsymbol{\lambda}, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$
- (d) $p(u | \boldsymbol{\lambda}) = \frac{\int_v p(u)p(v)p(\boldsymbol{\lambda}|u,v)}{\int_v \int_{\boldsymbol{\lambda}} p(u)p(v)p(\boldsymbol{\lambda}|u,v)}$

4. (15%) Suppose now that you also have access to covariates $\mathbf{x}_i$ that describe characteristics of the each individual i that could help explain his/her latent log-rate λ_i . You may assume that the relationship between the mean of λ_i and $\mathbf{x}_i$ is linear in parameters. Which of the following generative processes corresponds to this enhanced model?

Option 1:

- (a) Draw $v \sim \text{Inverse-Gamma}(1, 1)$
- (b) For each individual $i \in \{1, \dots, I\}$:
 - (i) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mathbf{x}_i, v)$
 - (ii) For each observation $n \in \{1, \dots, N_i\}$ of the individual i :
 - A. Draw observed count $y_{i,n} \sim \text{Poisson}(\exp(\lambda_i))$

Option 2:

- (a) Draw $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma \mathbf{I})$
- (b) Draw $v \sim \text{Inverse-Gamma}(1, 1)$
- (c) For each individual $i \in \{1, \dots, I\}$:
 - (i) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\boldsymbol{\beta}^T \mathbf{x}_i, v)$
 - (ii) For each observation $n \in \{1, \dots, N_i\}$ of the individual i :
 - A. Draw observed count $y_{i,n} \sim \text{Poisson}(\exp(\lambda_i))$

Option 3:

- (a) Draw $v \sim \text{Inverse-Gamma}(1, 1)$
- (b) For each individual $i \in \{1, \dots, I\}$:
 - (i) For each observation $n \in \{1, \dots, N_i\}$ of the individual i :
 - A. Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mathbf{x}_i, v)$
 - B. Draw observed count $y_{i,n} \sim \text{Poisson}(\exp(\lambda_i))$

Option 4:

- (a) Draw $v \sim \text{Inverse-Gamma}(1, 1)$
- (b) For each individual $i \in \{1, \dots, I\}$:
 - (i) For each individual characteristic d :
 - A. Draw $\beta_d \sim \mathcal{N}(0, \sigma^2)$
 - (ii) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\boldsymbol{\beta}^T \mathbf{x}_i, v)$
 - (iii) For each observation $n \in \{1, \dots, N_i\}$ of the individual i :
 - A. Draw observed count $y_{i,n} \sim \text{Poisson}(\exp(\lambda_i))$